

Ex.No:1: Simulation of FDMA in MATLAB

```
clc; clear; close all;  
fs=1000; t=0:1/fs:1;  
TBW=600; Users=3;
```

```
CBW=TBW/Users;
```

```
fc=[100 300 500];
```

```
m=sin(2*pi*10*t);
```

```
s1=m.*cos(2*pi*fc(1)*t);
```

```
s2=m.*cos(2*pi*fc(2)*t);
```

```
s3=m.*cos(2*pi*fc(3)*t);
```

```
subplot(2,1,1)
```

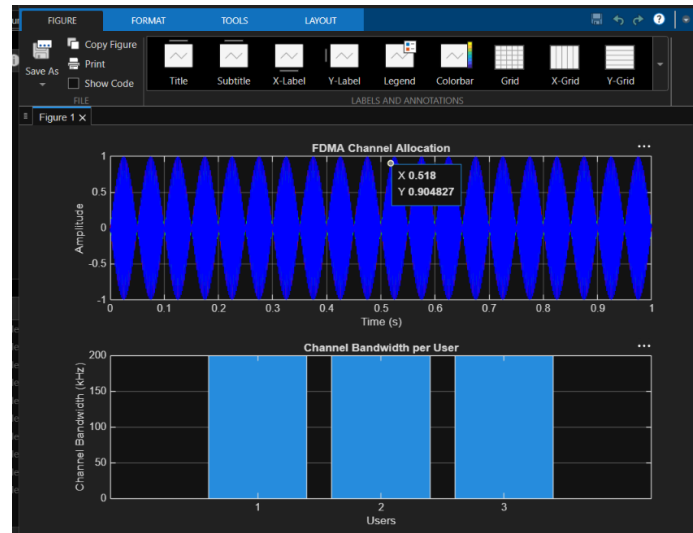
```
plot(t,s1,'r',t,s2,'g',t,s3,'b'),grid on
```

```
title('FDMA Channel Allocation'),xlabel('Time (s)'),ylabel('Amplitude')
```

```
subplot(2,1,2)
```

```
bar(1:Users,CBW*ones(1,Users)),grid on
```

```
xlabel('Users'),ylabel('Channel Bandwidth (kHz)'),title('Channel Bandwidth per User')
```



OBJECTIVE 2

```
clc; clear; close all;
```

```
BW = 200; % Channel Bandwidth (kHz)
```

```
Users = 3;
```

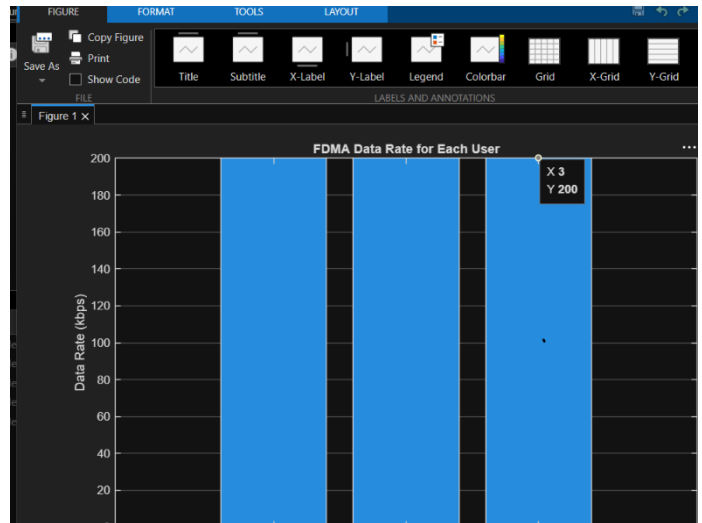
```
M = 2; % BPSK Modulation
```

```
DataRate = BW*log2(M); % Data Rate (kbps)
```

```

Rate = DataRate*ones(1,Users);
disp(' User Data Rate (kbps)')
disp([(1:Users)' Rate'])
figure
bar(1:Users,Rate)
xlabel('Users')
ylabel('Data Rate (kbps)')
title('FDMA Data Rate for Each User')
grid on

```



OBJECTIVE 3

```

clc; clear; close all;
Ps = 1; % Signal Power (W)
Pn = [0.1 0.05 0.01]; % Noise Power (W)
Users = 1:3;
SNR = 10*log10(Ps./Pn); % SNR in dB
disp('User Noise Power(W) SNR(dB)')
disp([Users' Pn' SNR'])
figure
bar(Users,SNR)
xlabel('Users')
ylabel('SNR (dB)')
title('Signal-to-Noise Ratio of FDMA Users')
grid on

```

